"Gravity must be caused by an agent acting constantly according to certain laws, but whether this agent be material or immaterial..." Isaac Newton, 1693 21 Oct 2025

The Foundation of Gravitation: Vectors, Angular Kinematics and Elements of General Relativity

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Abstract. This exposition examines how velocity direction, conservation laws and assumptions about the source determine gravitational motion. Circular acceleration and angular momentum lead to the calculation of orbital speeds, escape conditions, Keplerian periods and rendezvous timing. The distinction between gravitational potential and potential energy is developed through exterior and interior spherical models. The altitude law requires spherical symmetry, while the linear depth law additionally requires uniform density. A velocity-Verlet calculation of an eccentricity-0.6 orbit is compared with the analytic solution. At 20,000 steps per period, the maximum relative energy error is 7.31×10^{-7}. Halving the time step reduces that error by approximately a factor of four over the reported runs, providing a numerical consistency check. The relativistic discussion uses the supplied Schwarzschild exterior interval and radial geodesic energy constant to calculate supported-clock rates, radial light propagation and free-fall proper time. Finite falling-clock time to the horizon is distinguished from divergent Schwarzschild coordinate time, and coordinate light speed from locally measured light speed. Algebra, geometry, vectors and single-variable calculus suffice for the calculations presented. The Schwarzschild solution is not derived from the field equations. The manuscript synthesises established results and model calculations without claiming new physical laws or observational evidence.



Earthrise, Apollo 8, 24 December 1968. NASA/Bill Anders.

1. Introduction and scope

Give two bodies the same speed at the same distance from a planet. One may follow a circle and the other an ellipse, because their velocity directions differ. Energy alone cannot distinguish them. Angular momentum supplies the missing constraint. This elementary comparison sets the problem pursued here: what information is needed to calculate a gravitational trajectory, and what may be inferred from the motion once it is known?

Circular motion is a useful starting point because it separates change of direction from change of speed. The inward acceleration identifies the force required at an instant. Torque then determines whether angular momentum is conserved, while work determines how kinetic energy changes. Under an inverse-square central force, these relations constrain the size, shape and period of an orbit. An encounter requires the two bodies to occupy the same position at the same time. A completed orbital rendezvous also requires their velocities to match.

The force law also needs a model of its source. Outside a spherical body, only the total mass enters the field. Inside it, the enclosed mass changes with radius. Potential records the work per unit test mass, including contributions from shells whose net interior force vanishes. These distinctions become necessary both when calculating a path and when using a measured path to infer mass.

The Newtonian treatment uses a common time coordinate and Euclidean space. It does not provide a law for comparing clocks held at different gravitational potentials or carried along different relativistic paths. Those questions require an interval that relates coordinate changes to local measurements. The Schwarzschild exterior supplies the specific geometry used here. Its weak-field limit recovers a relation involving Newtonian potential, but potential and spacetime curvature remain different quantities.

The calculations assume algebra, geometry, vectors and single-variable differentiation and integration. Circular acceleration, the work–energy relation, spherical field and potential profiles, and several consequences of Keplerian motion are derived. The shell theorem, the conic form of inverse-square orbits, the Schwarzschild interval and the radial geodesic energy constant are supplied results. Worked applications test the use of these relations. A numerical orbit calculation checks a particular implementation of the Newtonian model. None of these calculations substitutes for deriving the relativistic field equations or testing a gravitational theory against observations.

1.1. Organisation of the analysis

Table 1: Organisation of the analysis and principal methods.

Section	**Subject**	**Principal method**
2	Circular motion and perceived gravity	Acceleration and force
3	Angular momentum and torque	Angular momentum conservation
4	Work and potential energy	Energy conservation
5	Circular orbits and escape	Force and energy
6	Ellipses and orbital phasing	Energy and angular momentum
7	Gravitational inference	Orbital measurements
8	Proper time and geometry	The spacetime interval

1.2. Mathematical conventions

A derivative is a local rate of change. Position changing with time gives velocity; velocity changing with time gives acceleration. Both velocity and acceleration have directions. Speed is only the magnitude of velocity.

$$v_x = \frac{dx}{dt}, \qquad a_x = \frac{dv_x}{dt} \tag{1}$$

An integral adds many small contributions. Work adds force along small displacements. A clock's elapsed proper time adds the small intervals that clock actually experiences.

$$W = \int F_x \, dx, \qquad \Delta\tau = \int \frac{d\tau}{dt} \, dt \tag{2}$$

For vectors $\mathbf{a}$ and $\mathbf{b}$, the dot product is $\mathbf{a} \cdot \mathbf{b} = ab\cos\theta$. It selects the component along the other vector. The cross product has magnitude $|\mathbf{a} \times \mathbf{b}| = ab\sin\theta$, so it includes the perpendicular lever arm. Its direction follows the right-hand rule. For vectors in the page, that direction is into or out of the page.

A radian is arc length divided by radius. One revolution is 2π radians. Radians are dimensionless, so $v = \omega r$ has units of metres per second.

Useful constants for the worked examples are g = 9.81 m s^{-2}, Earth's mean radius $R_E = 6.371 \times 10^6$ m, Earth's gravitational parameter $\mu_E = GM_E = 3.986 \times 10^{14}$ m^3 s^{-2}, and c = 2.998×10^8 m s^{-1}. These are rounded teaching values, not precision ephemerides.

For the solar-mass and kiloparsec examples, use $GM_\odot = 1.327 \times 10^{20}\ \mathrm{m^3\,s^{-2}}$ and $1\ \mathrm{kpc} = 3.086 \times 10^{19}\ \mathrm{m}$. These rounded conversion values suffice to reproduce the reported estimates. [22]

Notation. T means period; F_T means string tension. The symbol τ means torque in Section 3 and proper time in Section 8. Context distinguishes them. Lowercase h denotes altitude in Section 4 and specific angular momentum in orbital equations. Depth below the surface is d, with $r = R - d$. The symbol a means acceleration in Section 2 and semimajor axis in Section 6.

2. Circular motion and perceived gravity

2.1. Support at the rotating rim

Consider a person standing at the inner rim of a rotating space station, far from planets. Their path is circular even though they remain at rest relative to the floor. The floor must therefore push them inward. Their apparent weight is this support force, while the direction they call down points toward the rim.

Two arrows keep the descriptions separate. The contact force points toward the axis. The perceived direction of down points away from it. A force diagram must contain the actual contact force appropriate to the chosen frame.

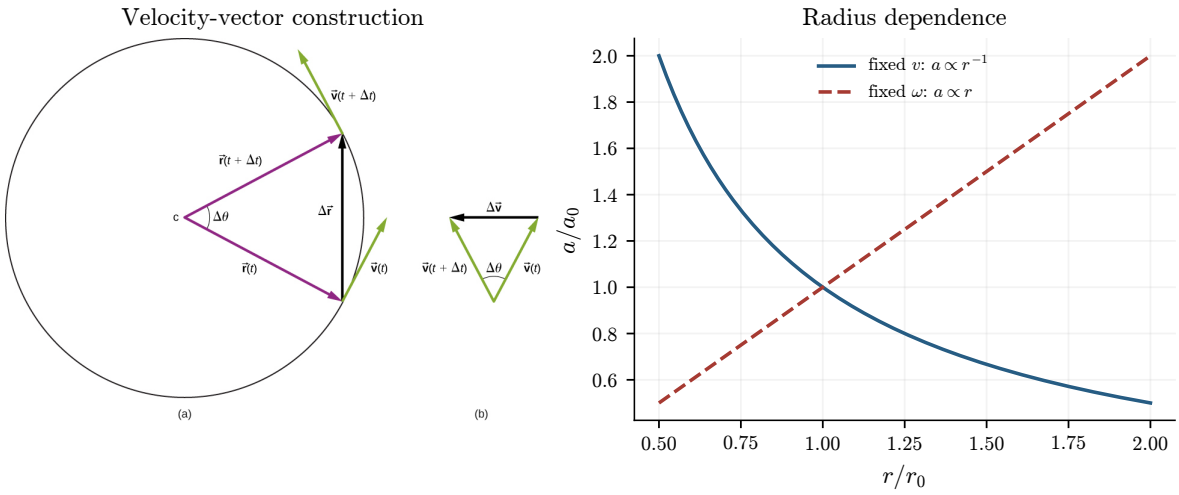


Figure 1. Circular motion. Left: position and velocity triangles illustrating the inward change of velocity (OpenStax, figure 4.18). Right: calculated acceleration as radius varies, with speed or angular speed held fixed; each curve is normalised at its reference radius. OpenStax diagrams: CC BY-NC-SA 4.0. Access for free at `https://openstax.org/books/university-physics-volume-1/pages/1-introduction` .

2.2. Angular speed and tangential velocity

At the rightmost point of a counterclockwise circle, velocity points upward. A quarter-turn later it points left. Even if the speedometer never changes, the velocity vector has

changed, so acceleration exists.

For radius r and period T, one circuit covers 2πr. Frequency f counts circuits per second, while angular speed ω counts radians per second.

$$f = \frac{1}{T}, \qquad \omega = \frac{2\pi}{T}, \qquad v = \frac{2\pi r}{T} = \omega r \tag{3}$$

All points of a rigid turntable share one angular speed ω. Their speeds v increase with distance from the axis. Independently orbiting bodies obey no such rigid constraint. Consequently, the radius dependence of v^2/r must be evaluated with the specified quantity held fixed.

2.3. Angular kinematics and nonuniform circular motion

An angular coordinate specifies position around a fixed axis. Its first time derivative is angular velocity and its second is angular acceleration. In planar motion a positive sign denotes counterclockwise rotation under the chosen convention. Their units are radians per second and radians per second squared.

$$\omega = \frac{d\theta}{dt}, \qquad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \tag{4}$$

For motion on a circle of fixed radius r, the signed tangential velocity is rω and its tangential acceleration is rα. The velocity direction also changes, producing an inward acceleration $r\omega^2$. With outward radial unit vector $\hat{\mathbf{r}}$ and counterclockwise tangent $\hat{\boldsymbol{\theta}}$, the complete planar vectors are

$$\mathbf{v} = r\omega\hat{\boldsymbol{\theta}}, \qquad \mathbf{a} = -r\omega^2\hat{\mathbf{r}} + r\alpha\hat{\boldsymbol{\theta}} \tag{5}$$

The two acceleration components are perpendicular, so the magnitude of the total acceleration is the square root of the sum of their squares. Uniform circular motion is the special case $\alpha = 0$; the radial component remains nonzero.

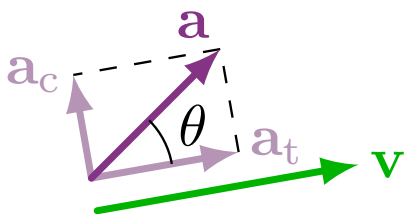


Figure 2. Acceleration decomposition by Izaak Neutelings [19]. The total acceleration is the vector sum of perpendicular tangential and normal components. The source labels the normal component a_c and the tangential component a_t. For a circle of fixed radius, their magnitudes are $r\omega^2$ and $r|\alpha|$. Reproduced as vector artwork under CC BY-SA 4.0.

When α is constant, integrating once gives the angular velocity and integrating again gives angular position. Eliminating time yields a third useful relation:

$$\omega = \omega_0 + \alpha t, \qquad \theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2 \tag{6}$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \tag{7}$$

These relations require a fixed axis, a consistent angular sign convention and constant angular acceleration. For example, a rotor starting from rest with $\alpha = 0.4$ rad s^{-2} reaches 2 rad s^{-1} after 5 s and turns through 5 radians. At radius 0.5 m its tangential acceleration is 0.2 m s^{-2} throughout, whereas its inward acceleration reaches 2 m s^{-2} at that final time. The distinct behaviour of the two components explains why a changing angular speed requires more than an inward-force condition alone.

2.4. Derivation of centripetal acceleration

Compare the velocity vectors at two nearby points on a circle. They have equal lengths and a small angle $\Delta\theta$ between them. The small triangle formed by their difference is similar to the triangle formed by the two radius vectors.

$$\frac{|\Delta \mathbf{v}|}{v} \simeq \frac{\Delta s}{r} \tag{8}$$

Since Δs/Δt approaches v, divide by Δt and take the limit. The change in velocity points toward the centre in that limit.

$$a_c = \frac{v^2}{r} = \omega^2 r \tag{9}$$

The result can be checked using single-variable calculus. Write the circular trajectory as x = r cos(ωt), y = r sin(ωt). Differentiate each coordinate twice. Differentiation gives $a_x = -\omega^2$x and $a_y = -\omega^2$y: acceleration is the negative of the radius vector multiplied by ω^2.

If the speed changes as well, acceleration has an additional tangential component. The inward component v^2/r is still present. Uniform circular motion is the special case with zero tangential component.

2.5. Forces producing circular motion

The inward component of the net force must supply this acceleration. Tension, gravity, friction or a normal force may contribute. The term centripetal names the required direction of the resultant. It does not identify an additional interaction.

$$\sum F_{\text{inward}} = \frac{mv^2}{r} \tag{10}$$

Cutting a string removes its tension without changing the instantaneous velocity. The object initially moves along the tangent. Its subsequent path is straight if no net force remains, or curved if gravity or another force continues to act.

2.6. A habitat with specified apparent weight

Take a habitat of radius 100 m. Require an inward acceleration of one Earth g at its rim. Ignore the station's own gravity and the occupant's motion relative to the floor.

$$\omega = \sqrt{\frac{g}{r}} = 0.313 \text{ rad s}^{-1} \tag{11}$$

$$T = \frac{2\pi}{\omega} = 20.1 \text{ s}, \qquad v = \sqrt{gr} = 31.3 \text{ m s}^{-1} \tag{12}$$

The rotation rate is about 2.99 revolutions per minute. For a person stationary relative to the rim, the floor supplies $N = mg$ inward. The symbol g specifies the desired acceleration in this calculation. The station has produced a support force without a planetary gravitational field.

At the same angular speed, the required acceleration is proportional to distance from the axis. At 50 m it is half the rim value; at the axis it is zero. Moving around inside a rotating habitat introduces additional effects, including Coriolis deflection, beyond this stationary-occupant calculation.

If the radius is doubled while the required rim acceleration stays fixed, the angular speed falls by $\sqrt{2}$, the period grows by $\sqrt{2}$, and the rim speed grows by $\sqrt{2}$. The constraint is fixed apparent weight, not fixed rotation rate.

2.7. Maintaining contact in a vertical circle

At the top of a vertical circle, toward the centre is downward. Gravity points downward, and the bottom of the upside-down bucket can also push the water downward. Let N be that contact force on a representative bit of water.

$$mg + N = \frac{mv_{\text{top}}^2}{r} \tag{13}$$

For the stated contact model, N cannot be negative. Setting $N = 0$ identifies the threshold at which gravity alone supplies the inward acceleration.

$$v_{\text{top}} \geq \sqrt{gr} \tag{14}$$

For r = 1 m, the minimum top speed is 3.13 m s^{-1}. Gravity supplies the entire inward acceleration at that threshold. The result is a condition on the speed **at the top**, not an instruction to use one constant speed everywhere.

If a point mass on a taut, massless string passes through a complete vertical loop with only gravity doing work, the bottom is 2r lower. Energy conservation gives $v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr$. At the limiting top speed:

$$v_{\text{bottom}} = \sqrt{5gr}, \qquad F_{T,\text{bottom}} = 6mg \tag{15}$$

The vanishing contact force at the top therefore coexists with a large tension at the bottom. Force balance gives the condition at each location. The relation between the two speeds comes from the work done during the descent, developed in Section 4.

2.8. Acceleration in a centrifuge

At 1,200 rpm, frequency is 20 s^{-1} and $\omega = 40\pi$ rad s^{-1}. At radius 0.15 m:

$$a_c = (40\pi)^2(0.15) = 2.37 \times 10^3 \text{ m s}^{-2} \simeq 241g \tag{16}$$

The result specifies the required inward force per unit mass and the corresponding effective acceleration in the rotating frame. The particle's mass is unchanged. In a liquid, separation also depends on density difference, drag, particle size and thermal motion, so the circular point-particle model supplies only part of the description.

3. Angular momentum and torque

3.1. What contraction preserves

A skater pulling in their arms can increase both angular speed and rotational kinetic energy while the external torque remains negligible. The preceding force calculation is not enough to explain this change. We must identify the conserved quantity and distinguish it from the energy transferred during contraction.

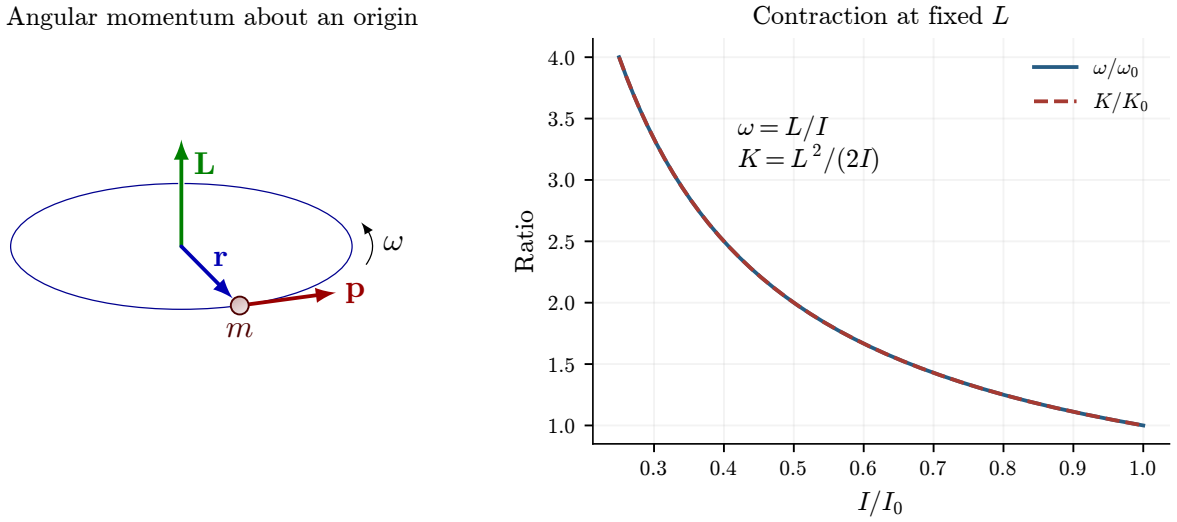


Figure 3. Angular momentum and contraction. Left: a particle in circular motion, with position **r**, tangential momentum **p** and perpendicular angular momentum **L** following the right-hand rule (Izaak Neutelings [19]). The circle appears elliptical in perspective. Right: calculated angular speed and rotational kinetic energy at fixed angular momentum; the normalised curves coincide. Illustration: CC BY-SA 4.0.

3.2. Dependence on the chosen origin

Choose a fixed point O in an inertial frame. For a particle, r points from O to the particle and p = mv is its momentum.

$$\mathbf{L}_O = \mathbf{r} \times \mathbf{p}, \qquad |L_O| = mvr \sin\theta = mvb \tag{17}$$

Here b is the perpendicular distance from O to the particle's line of motion. A particle moving along a straight line can have angular momentum about a point away from that line. It does not need to follow a circle.

For a particle moving left above O, $\mathbf{r} \times \mathbf{p}$ points out of the page. If O is moved onto the line of motion, the perpendicular distance b becomes zero. The motion is unchanged, but its angular momentum about the new origin is zero.

Linear momentum also depends on the reference frame. In a chosen inertial frame it does not additionally require an origin; angular momentum does.

3.3. The torque equation

Differentiate r × p about that fixed inertial origin. The product rule gives two terms. Since dr/dt = v and p = mv, v × p is zero.

$$\frac{d\mathbf{L}_O}{dt} = \mathbf{v} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} = \mathbf{r} \times \mathbf{F} = \boldsymbol{\tau}_O \tag{18}$$

The torque magnitude is Fb, where b is now the perpendicular distance to the force's line of action. A push along the line from a hinge to a door handle has no moment about the hinge. A transverse push has a moment set by that perpendicular distance.

Torque and angular momentum must use the same origin. Zero torque about the centre of a planet does not automatically mean zero torque about some other point.

3.4. Angular momentum under a central force

Gravity from an ideal spherical central body points along the radius. Parallel or antiparallel vectors have zero cross product, so r × F = 0 about that centre. Angular momentum about the centre is conserved even on an ellipse, where speed and distance change continuously.

For a system of particles, the internal pair torques cancel when the pair forces are equal, opposite and directed along the line joining the particles. Equal and opposite forces with different lines of action can instead form a couple. With the usual central internal forces of the Newtonian particle model, the remaining torque is the total external torque.

$$\frac{d\mathbf{L}_{\text{total}}}{dt} = \boldsymbol{\tau}_{\text{external}} \tag{19}$$

Zero external torque fixes the total vector, not necessarily each object's share of it. Components within a system can exchange angular momentum.

3.5. Moment of inertia and axial rotation

For rotation about a fixed symmetry axis, a mass m_i at perpendicular distance ρ_i has speed $\omega\rho_i$. Its axial angular momentum is $m_i\rho_i^2\omega$. Summing over the masses defines the

moment of inertia.

$$I = \sum_i m_i \rho_i^2, \qquad L_{\text{axis}} = I\omega \tag{20}$$

Moment of inertia specifies the mass distribution relative to the axis. Equal total mass does not mean equal I. Moving mass twice as far from an axis multiplies that part's contribution by four.

The vector statement L = Iω needs the appropriate principal or symmetry axis. We do not use it as a universal rule for arbitrary three-dimensional rotation; that general problem requires more machinery than this paper assumes.

3.6. Spin and orbital contributions

A spinning satellite can orbit Earth. About Earth's centre, one contribution comes from the centre of mass going around Earth, and another from rotation about the satellite's own centre of mass.

$$\mathbf{L}_O = \mathbf{R}_{\text{cm}} \times \mathbf{P} + \mathbf{L}_{\text{cm}} \tag{21}$$

The centre-of-mass term is orbital angular momentum about O. The second term is spin angular momentum about the centre of mass. Both arise from the same definition, applied to different parts of the total motion.

3.7. The work required for contraction

Let I change from 6 to 2 kg m^2. Initially $\omega = 2$ rad s^{-1}. With negligible external torque, L = 12 kg m^2 s^{-1} throughout, so the final angular speed is 6 rad s^{-1}.

Rotational kinetic energy is K = $I\omega^2/2$. Substitute ω = L/I to see what fixed L implies.

$$K_{\text{rot}} = \frac{L^2}{2I} \tag{22}$$

The initial energy is 12 J and the final energy is 36 J. The additional 24 J is supplied by work during contraction, drawn from the skater's internal energy. The torque condition fixes angular momentum while leaving room for this energy transfer.

Holding ω fixed with an external motor changes the problem. Angular momentum then changes with I, and the motor or support exchanges torque with the system. The energy calculation must follow that new constraint.

3.8. A contracting stellar core

Model a collapsing stellar core as I = kMR^2. If its mass M, structural factor k, and angular momentum remain unchanged, then ω is proportional to $1/R^2$ and the rotation period is proportional to R^2.

$$\frac{T_f}{T_i} = \left(\frac{R_f}{R_i}\right)^2 \tag{23}$$

A hypothetical core shrinking from 10,000 km to 10 km shrinks in radius by 1,000, so its period shrinks by one million. An initial 30-day period becomes about 2.59 seconds.

This estimate isolates the effect of contraction at fixed mass, structural factor and angular momentum. Real collapse can eject mass and angular momentum, change the density distribution and require relativity. The hypothetical core therefore illustrates a mechanism without predicting the final state of a particular star.

Pulsars are rapidly rotating neutron stars whose emission can sweep past us. If the beam crosses our line of sight, rotation produces pulses. Beam geometry matters: not every neutron star presents a beam to Earth.

3.9. Limits of the compact-remnant comparison

For compact stellar remnants, support and gravity determine what configurations are possible, while angular momentum constrains rotation during change. The detailed mass thresholds and equations of state are beyond this section.

For a black hole of nonzero mass, the horizon has a finite scale. It is distinct from the classical central singularity. A uniform material density does not define a black hole, so the preceding contraction model cannot be extrapolated into a description of its interior. Section 8 specifies the exterior model used for the relativistic calculation.

4. Work, energy and gravitational potential

4.1. The meaning of negative mechanical energy

Angular momentum constrains turning, but it does not decide whether a body can escape to arbitrarily large radius. For that purpose we need the work done along its path and a reference for potential energy. With zero potential energy at infinity, a negative total mechanical energy expresses a restriction on the accessible motion.

4.2. Definition of mechanical work

A force perpendicular to the instantaneous displacement does no instantaneous work. A force along motion adds kinetic energy; a component opposite motion removes it. In one dimension, add the small contributions F_x dx.

$$W_{A\to B} = \int_{x_A}^{x_B} F_x \, dx \tag{24}$$

For a constant force making a fixed angle θ to a straight displacement of length d, W = Fd cos θ. In a curved trajectory, apply that projection to each small segment rather than treating the whole path as one displacement.

The more compact notation is the dot-product integral $\mathrm{W} = \int \mathrm{F} \cdot \mathrm{dr}$. Multivariable integration is not required for the path calculation here: parameterise a path by its one time variable and integrate F · v with respect to time.

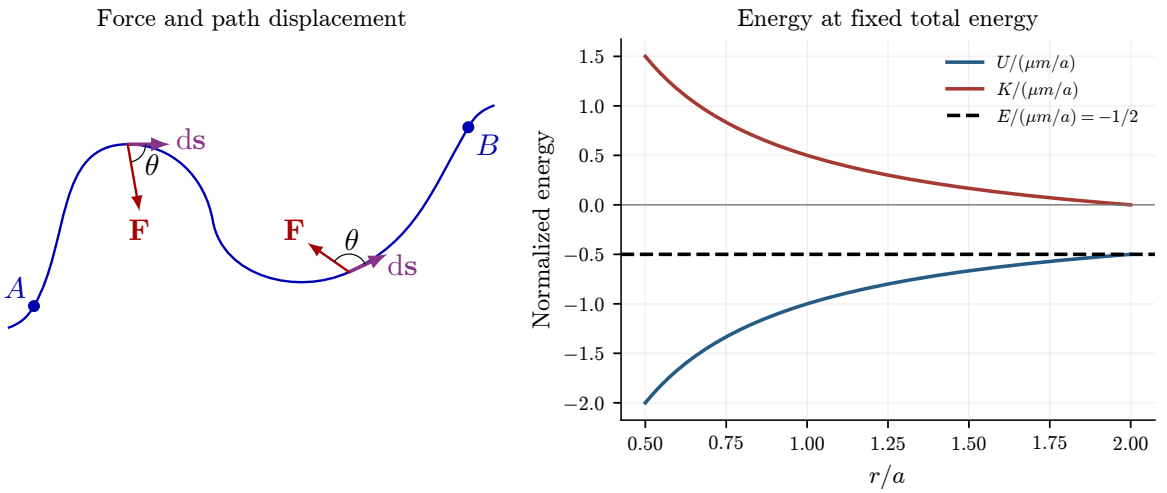


Figure 4. Work and energy. Left: the path diagram shows force and tangential infinitesimal displacement at two points on a curved path (Izaak Neutelings [19]). The source writes the displacement as d**s**, equivalent to d**r** here. The angle θ determines the signed work element $F\cos\theta\,\mathrm{d}s$. Right: potential and kinetic energy at fixed total energy $E = -\mu m/(2a)$. The curves do not specify the allowed radial interval of a particular nonradial orbit without its angular momentum. Illustration: CC BY-SA 4.0.

4.3. Derivation of the work energy theorem

For constant mass in one dimension, use $F = m\,dv/dt$ and $dx = v\,dt$. Substitution changes the work integral into an integral over velocity.

$$W = \int m\frac{dv}{dt}v\,dt = \int mv\,dv = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 \tag{25}$$

The result concerns the work of the net force. The work of one selected force gives the total kinetic-energy change only when every other force does zero net work.

In uniform circular motion, the net inward force is perpendicular to velocity, so its power F · v is zero. It turns the velocity without changing its magnitude. Force and work answer different questions.

4.4. Work by an oblique force

A 2 kg cart initially moves at 3 m s^{-1}. A constant 30 N force acts at 60 degrees to a 5 m horizontal displacement. Assume other forces do no net work over that motion.

$$W = (30)(5)\cos 60^\circ = 75\ \mathrm{J} \tag{26}$$

Initial kinetic energy is 9 J, so final kinetic energy is 84 J. Since K = $mv^2/2$ and m = 2 kg, the final speed is $\sqrt{84} = 9.17$ m s^{-1}. The full 30 N is not the component doing horizontal work.

4.5. Conservative forces and potential energy

For gravity in this model, the work between two locations does not depend on the path. We define potential energy U so that the work done by gravity is minus the change in U.

$$W_{\text{gravity}} = -\Delta U \tag{27}$$

With gravity as the only force doing work, ΔK = $-\Delta$U, so K + U is constant. The same logic works for an ideal spring. In one dimension, force and potential are connected by $F_x = -\text{dU/dx}$.

Friction generally does path-dependent work, and mechanical energy can turn into thermal energy. If a block returns to its original position after sliding around a rough loop, friction has still removed mechanical energy. A single-valued position potential cannot account for that loss.

For the chosen mechanical system, one useful accounting form is $K_B + U_B = K_A + U_A + W_{other}$. Forces already represented by U must not also be counted again in W_{other}.

4.6. The reference value of potential energy

Near Earth's surface, with nearly constant g and vertical coordinate z increasing upward, U = mgz + C. Changing C changes the number assigned to U everywhere by the same amount. It does not change force or energy differences.

Far from the surface, the force weakens. Write μ = GM and let r increase outward. The radial gravitational force is $-\mu\text{m}/r^2$. Define U = 0 at infinity and integrate its negative.

$$U(r) - U(\infty) = -\int_{\infty}^{r} \left(-\frac{\mu m}{s^2}\right) ds = -\frac{\mu m}{r} \tag{28}$$

As attracting masses are brought together from infinity, gravity does positive work and the potential energy decreases. Its negative value records the work needed to separate them again to the chosen reference. Changing the reference shifts the assigned energy without changing that required work.

4.7. Lifting through a changing field

Move a mass m slowly from radius R to 2R, outside a spherical Earth. Since initial and final speeds are negligible, the work supplied by the lifter equals the increase in U.

$$\Delta U = -\frac{\mu m}{2R} + \frac{\mu m}{R} = \frac{\mu m}{2R} \tag{29}$$

Using surface g times the full height R would give mgR = μm/R, twice the correct value. The constant-g approximation fails over a height comparable to Earth's radius.

For a small height z above the surface, the exact difference is:

$$U(R+z) - U(R) = \frac{\mu m z}{R(R+z)} \simeq mgz \quad (z \ll R) \tag{30}$$

The constant-g expression follows when the change in radius is small compared with R. Its range of validity is therefore part of the calculation, rather than a separate rule to remember.

4.8. Gravitational potential and potential energy

Potential energy belongs to a specified test mass in a specified gravitational field. Gravitational potential removes the test-mass factor and describes the energy assigned per unit mass at a position. The test mass is assumed small enough that it does not appreciably alter the source. The distinction is

$$\Phi(r) = \frac{U(r)}{m}, \qquad U(r) = m\Phi(r), \qquad [\Phi] = \mathrm{J\,kg^{-1}} = \mathrm{m^2\,s^{-2}} \tag{31}$$

Doubling the test mass at one position doubles U while leaving Φ unchanged. Potential is a scalar with units of energy per unit mass. The gravitational field is a vector with units of force per unit mass. The source determines both, but they enter the calculation in different ways. [15, 16]

For a slow displacement from A to B, assume gravity and the external agent are the only forces doing work, with negligible change in kinetic energy. The external work is then the increase in potential energy. Dividing by the test mass gives the potential difference. In the spherical case, differentiation gives the signed radial force and field.

$$\Phi_B - \Phi_A = \frac{W_{\mathrm{ext}}}{m} = -\frac{W_{\mathrm{gravity}}}{m}, \qquad F_r = -m\frac{d\Phi}{dr}, \qquad g_r = -\frac{d\Phi}{dr} \tag{32}$$

Here the radial coordinate increases outward and the signed radial field component is g_r. The positive magnitude is written $g = |g_r|$. In the attractive spherical models considered below, $g_r = -g$ and consequently $d\Phi/dr = g$. The field depends on the slope of the potential, rather than its absolute value. Adding a constant to potential changes neither force nor motion.

For a point source, or outside a spherically symmetric source of total mass M, set the potential to zero at infinity. Integrating the exterior field gives

$$\Phi(r) - \Phi(\infty) = \int_{\infty}^{r} \frac{GM}{s^2}\, ds = -\frac{GM}{r}, \qquad \mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}} \tag{33}$$

Moving outward makes the potential less negative. Moving along a sphere centred on the source leaves it unchanged. Such a sphere is an equipotential surface, and gravity does no work along it. The circular-orbit result is therefore also a consequence of the potential description.

Using Earth's gravitational parameter and mean radius quoted earlier, the ideal spherical surface potential is approximately -6.26×10^7 J kg^{-1}. A 2 kg test mass there has potential energy approximately -1.25×10^8 J with the infinity reference. These large negative values do not imply that a small lift requires that much work. The work of a small lift depends on the potential difference between its two endpoints.

4.9. Variation of gravitational acceleration with altitude

Let R be the source's surface radius, let h be altitude above that surface, and define the ideal surface field magnitude $g_0 = GM/R^2$. The distance entering the inverse-square law is the distance from the centre, $r = R + h$, rather than the altitude alone. For a spherical source with all of its mass below the surface,

$$g(h) = \frac{GM}{(R+h)^2} = g_0 \left(1 + \frac{h}{R}\right)^{-2} \tag{34}$$

The exterior shell theorem makes the altitude law independent of the interior density profile, provided the mass distribution is spherical. Rotation, nonsphericity and other bodies have been omitted. The result describes gravitational acceleration. A scale on a rotating Earth also responds to the support needed to maintain its motion.

At small altitude, the derivative of $(1 + x)^{-2}$ at $x = 0$ is -2. Retaining only the constant and linear terms gives

$$g(h) \simeq g_0 \left(1 - \frac{2h}{R}\right), \qquad \frac{g(h) - g_0}{g_0} \simeq -\frac{2h}{R} \quad (h \ll R) \tag{35}$$

The omitted terms begin at second order in h/R. Extending the linear expression to $h = R/2$ would give zero, although the exact exterior law remains positive at every finite radius. That failure marks the end of the approximation's useful range.

At 400 km altitude, with $R = 6,371$ km, the exact ratio is $g(h)/g_0 \simeq 0.8853$. The linear approximation gives 0.8744, so it underestimates the exact field by about 1.23 percent. Gravity at a low-orbiting station is therefore still substantial. The station's apparent weightlessness follows from free fall. To reduce the gravitational acceleration to half its surface magnitude requires

$$g(h) = \frac{g_0}{2} \quad \Longrightarrow \quad h = (\sqrt{2} - 1)R \simeq 2,639 \text{ km} \tag{36}$$

4.10. Shell theorem and the variation with depth

Depth d below the surface corresponds to $r = R - d$. An interior point cannot be analysed by placing the entire source mass at its centre. The shell theorem states that a spherically symmetric shell produces zero net field at every interior point, while the field outside the shell is the same as that of its mass at the centre. Consequently, only the mass enclosed within radius r contributes to the net field there. This result supplies the physical step needed for the interior calculation. [15]

The shell's cancellation depends on the inverse-square law and the complete distribution of its mass. At an off-centre point, corresponding near and far portions have different distances and masses. Their integrated force contributions cancel. Their potentials add, which is why exterior shells must still be retained in an interior potential calculation.

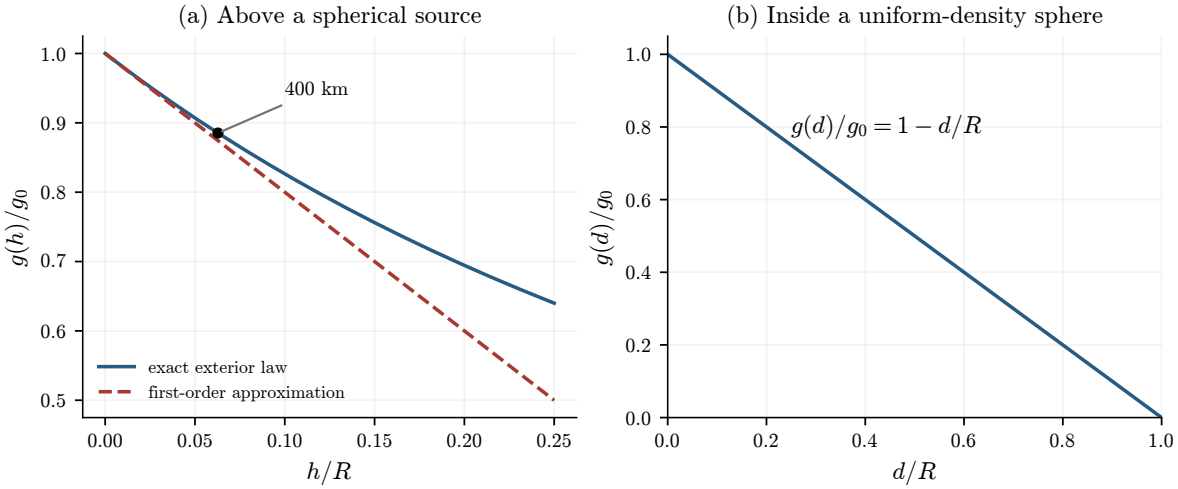


Figure 5. Calculated field variation for a spherical body. Left: the exact exterior altitude relation and its first-order approximation; the 400 km example is marked using Earth's mean radius. Right: the exact linear relation with depth for a uniform-density sphere. The two panels use different assumptions about the source interior. The curves are Python calculations, not measurements of Earth's interior.

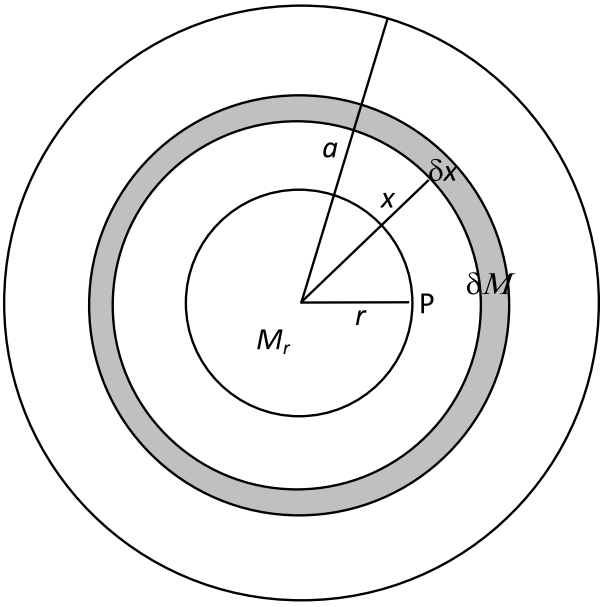


Figure 6. Concentric-shell construction from Jeremy Tatum, Celestial Mechanics. The vector artwork is cropped from figure V.37 of the author's chapter 5 PDF; the same construction appears as figure V.24A in the LibreTexts edition [15, 16]. CC BY-NC 4.0. The source labels the body's radius a, which is R here. At the point P, the enclosed mass is labelled M_r; the shaded shell has radius x and mass δM. An exterior shell contributes zero net field at P and a constant negative potential. Source: `https://phys.libretexts.org/@go/page/8153` .

For a uniform-density sphere, the density is $\rho = 3M/(4\pi R^3)$. The enclosed mass decreases with the cube of radius, and the interior field is

$$M(r) = \frac{4\pi}{3}\rho r^3 = M\left(\frac{r}{R}\right)^3, \qquad g(r) = \frac{GM(r)}{r^2} = \frac{GM}{R^3}r = g_0\frac{r}{R} \tag{37}$$

Expressing the same result in terms of depth gives the familiar formula

$$g(d) = g_0\left(1 - \frac{d}{R}\right), \qquad 0 \le d \le R \quad \text{(uniform density)} \tag{38}$$

For this model, gravity has its maximum magnitude at the surface, falls to half that value at depth $R/2$, and vanishes at the centre by symmetry. For example, a depth of 1,000 km gives $g(d)/g_0 \simeq 0.8430$. These numerical ratios describe the uniform-density model. They must not be interpreted as predictions from a realistic terrestrial density profile.

The change from altitude to depth changes the source term as well as the distance. Outside, all the mass is enclosed. Inside, only $M(r)$ contributes to the net field. Replacing h by $-d$ in $GM/(R+h)^2$ would keep the full mass incorrectly and predict a divergence at the centre.

4.11. Gravitational potential inside a uniform sphere

The exterior expression $-GM/r$ also cannot be continued into a solid sphere while keeping M fixed. At the surface the potential is $-GM/R$. Inside a uniform sphere, the field magnitude found above gives its radial derivative. Integrating inward from the known surface value determines the interior potential and its integration constant:

$$\Phi(r) - \Phi(R) = \int_R^r \frac{GM}{R^3}s\,ds = \frac{GM}{2R^3}(r^2 - R^2) \tag{39}$$

$$\Phi(r) = -\frac{GM}{2R^3}(3R^2 - r^2), \qquad 0 \le r \le R \tag{40}$$

Both potential and its first radial derivative match the exterior solution at the surface. The potential is a quadratic function inside and approaches zero as $-GM/r$ outside. At the centre,

$$g(0) = 0, \qquad \Phi(0) = -\frac{3GM}{2R}, \qquad \Phi(R) = -\frac{GM}{R} \tag{41}$$

Zero field means that the local potential slope is zero. It does not mean that the potential itself is zero. With the infinity reference, the centre is the minimum of the potential and has a more negative value than the surface. A test mass moving slowly from the centre to the surface requires positive external work $mGM/(2R)$.

Each shell outside P contributes a constant negative potential at P while supplying zero net field there. Dropping these constants can leave a force calculation intact but spoil the work calculated between two positions. Matching the integrated interior potential to its known surface value retains the missing contributions. [16]

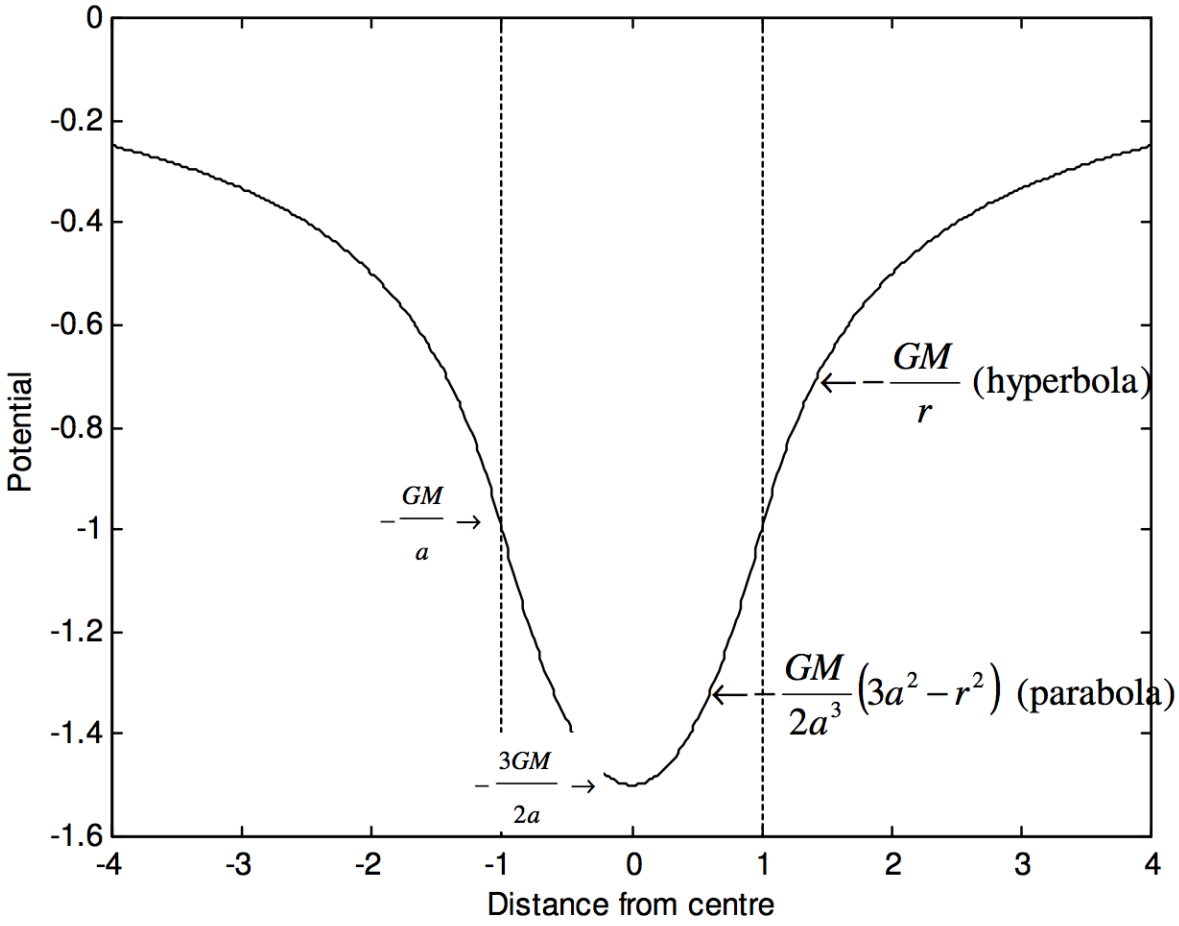


Figure 7. Interior and exterior potential profile from Jeremy Tatum, Celestial Mechanics, figure V.25, reproduced via Physics LibreTexts under CC BY-NC 4.0. The horizontal axis represents signed position along a diameter, in units of the source radius a, which is R here; physical radius is its absolute value. The vertical values are potential in units of GM/R. The centre has value -1.5, the surface -1, and infinity 0. The line is a theoretical profile, not measured data. Source: `https://phys.libretexts.org/@go/page/8153` .

4.12. Nonuniform density and the limits of the depth formula

For a spherically symmetric body with radial density $\rho(s)$, the enclosed mass must instead be calculated from its density profile. A thin spherical layer has mass $4\pi s^2\rho(s)\,ds$, so ordinary one-variable integration gives

$$M(r) = 4\pi \int_0^r \rho(s) s^2\, ds, \qquad g(r) = \frac{GM(r)}{r^2} \tag{42}$$

The corresponding interior potential includes the enclosed mass and every exterior shell:

$$\Phi(r) = -G\left[\frac{M(r)}{r} + 4\pi \int_r^R \rho(s) s\, ds\right] \tag{43}$$

These formulas reduce to the preceding uniform-sphere results when density is constant. At the centre their limiting values must be used. For a finite central density,

the enclosed mass is proportional to r^3 at small radius, the field tends to zero, and the potential remains finite.

A useful diagnostic follows by differentiating the field with respect to outward radius and defining the mean enclosed density $\bar{\rho}(r) = 3M(r)/(4\pi r^3)$:

$$\frac{dg}{dr} = 4\pi G\rho(r) - \frac{2GM(r)}{r^3} = 4\pi G\left[\rho(r) - \frac{2}{3}\bar{\rho}(r)\right] \tag{44}$$

Entering the body means decreasing r. The field magnitude therefore rises inward where the local density is less than two-thirds of the mean enclosed density. The decreasing distance then outweighs the loss of enclosed mass. Uniform density gives the opposite trend because local and mean density are equal.

Klotz's analysis of the gravity tunnel uses a seismically constrained terrestrial density profile rather than a uniform sphere. Its reconstructed field rises below the surface to about 1.09 times the surface magnitude before declining through the core to zero at the centre. The paper therefore provides a concrete example of why the linear depth law is a model result, rather than a universal law for the Earth. [17]

Both the uniform and nonuniform profiles remain Newtonian spherical test-mass models. Precision calculations require rotation, latitude, topography and local density anomalies to be treated separately. The density profile supplies information that the exterior mass and radius alone cannot provide.

4.13. A pendulum returning to its release height

Release a pendulum from rest at height z above its lowest position. In the ideal model, tension is perpendicular to motion and does no work; only gravity changes kinetic energy. At the bottom:

$$mgz = \frac{1}{2}mv^2, \qquad v = \sqrt{2gz} \tag{45}$$

At the opposite turning point, speed is zero again, so height is the same as at release. Real dissipation lowers the return height. An extra initial push would add energy and change the prediction.

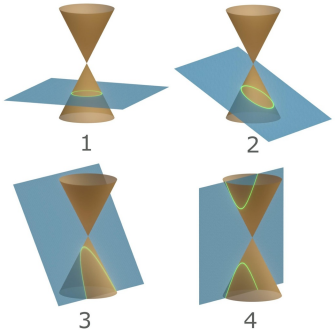
The equal return height follows only while the support remains fixed and no dissipative or driving work is added.

5. Circular orbits, escape and power

5.1. Free fall with a fixed orbital radius

A circular orbit applies the same inward-force relation as the rotating habitat, but gravity now supplies the force. An astronaut and the surrounding spacecraft can fall together with almost no contact force between them. Calculating the gravitational acceleration and calculating the scale reading are therefore separate tasks.

(a) Conic sections



(b) Quadratic drag and power

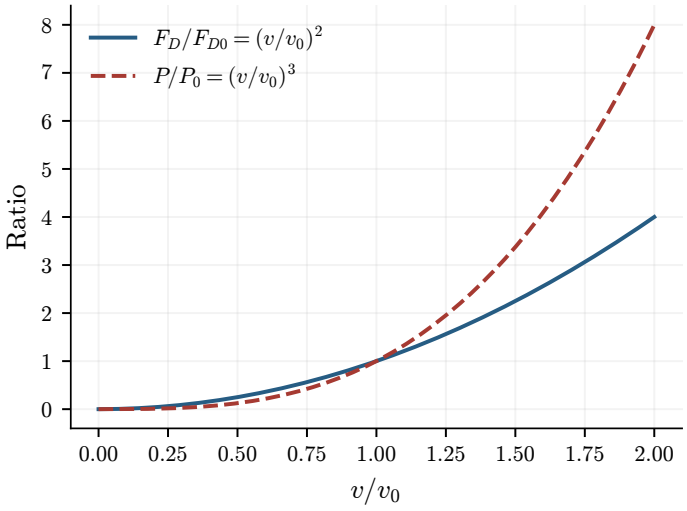


Figure 8. Conic geometry and power. Left: conic sections by JensVyff [20]: (1) circle, (2) ellipse, (3) parabola and (4) hyperbola. This geometrical illustration is not a scale plot of orbits. Right: quadratic-drag and power curves for a fixed drag coefficient, normalised at v_0. Illustration: CC BY-SA 4.0.

5.2. Dynamics of a circular orbit

For a circular orbit of radius r around a much more massive spherical body, set gravitational force equal to the required inward net force. The satellite mass cancels.

$$\frac{GMm}{r^2} = \frac{mv_c^2}{r}, \qquad v_c = \sqrt{\frac{\mu}{r}} \tag{46}$$

The satellite continuously falls toward the central body while its sideways motion carries it around. Its distance from the centre remains constant because its velocity keeps turning. Gravity supplies the inward acceleration; there is no balancing outward real force in the inertial description.

Inside a freely falling spacecraft, astronaut and spacecraft accelerate together. The astronaut does not need a floor force to follow the same local fall, so a scale can read nearly zero. Gravity remains substantial at low Earth orbit. Small tidal differences and other effects make the situation an approximation, not perfect absence of every force.

The orbital period follows from circumference divided by speed.

$$T = 2\pi\sqrt{\frac{r^3}{\mu}}, \qquad T^2 = \frac{4\pi^2 r^3}{\mu} \tag{47}$$

5.3. An orbit 400 km above Earth

The radius needed in the force law is measured from Earth's centre. An altitude of 400 km gives r = 6.771×10^6 m, not 400,000 m.

Using $\mu_E = 3.986 \times 10^{14}\ m^3\ s^{-2}$ gives $v_c = 7.67$ km s^{-1} and T = 92.4 minutes. The gravitational acceleration there is $\mu_E/r^2 \simeq 8.69$ m s^{-2}, about 89% of the surface value.

The field is still strong at this altitude. The small scale reading follows from the shared fall of astronaut and spacecraft.

5.4. Energy classification of trajectories

For inverse-square gravity with U(∞) = 0, total mechanical energy is:

$$E = \frac{1}{2}mv^2 - \frac{\mu m}{r} \tag{48}$$

At infinity, potential energy approaches zero and kinetic energy cannot be negative. Therefore an isolated trajectory with E < 0 cannot reach infinity. In an ideal point-mass problem with nonzero angular momentum it is an ellipse; a finite central body's surface can interrupt that mathematical orbit.

At E = 0, the limiting outward trajectory reaches arbitrarily large distance with speed approaching zero. It does not arrive at infinity at a finite time and then fall back. For nonzero angular momentum its orbit is parabolic.

At E > 0, an outward unbound trajectory retains speed v_∞, with $E = mv_\infty^2/2$. The nonradial orbit is hyperbolic. Positive energy does not prevent an object aimed at a planet from colliding with it first.

The classification applies after specifying the initial state and excluding thrust, drag, collisions and major third-body perturbations. Energy determines whether infinity is energetically accessible. It does not establish clearance from a finite central body.

5.5. Derivation of escape speed

Set the energy to the escape threshold, E = 0, at launch radius r. The satellite mass cancels once more.

$$\frac{1}{2}mv_{\text{esc}}^2 = \frac{\mu m}{r}, \qquad v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} = \sqrt{2}\, v_c \tag{49}$$

From Earth's surface this ideal value is about 11.2 km s^{-1}. It ignores atmosphere and rotation. It is an initial coast speed for the model, not a requirement that an accelerating rocket instantly attain this speed while still on the launch pad.

Gravity has not stopped acting after escape. It continues to change the trajectory, but the object has sufficient total energy to remain unbound.

5.6. The impulse from circular motion to escape

At 400 km altitude, $v_c = 7.67$ km s^{-1}, so the local escape speed is 10.85 km s^{-1}. An ideal instantaneous prograde impulse therefore needs:

$$\Delta v = (\sqrt{2} - 1)v_c \simeq 3.18 \text{ km s}^{-1} \tag{50}$$

The difference is a vector change here because the impulse is explicitly parallel to the original velocity. A differently directed impulse with the same magnitude need not produce the same final speed or energy.

5.7. Energy and speed in circular orbits

In a circular orbit, substitute $v_c^2 = \mu/r$ into K and E.

$$K = \frac{\mu m}{2r}, \qquad U = -\frac{\mu m}{r}, \qquad E = -\frac{\mu m}{2r} \tag{51}$$

As r increases, E becomes less negative, so energy has increased. Meanwhile K decreases. The potential-energy increase is larger than the kinetic-energy decrease.

A prograde burn first increases speed at the burn location. The resulting path is generally a new ellipse. During the outward motion, gravity transfers kinetic energy into potential energy, and a second burn is usually needed to circularise at the larger radius. The speed immediately after the first burn and the speed in the final circular orbit refer to different events.

5.8. Mechanical power and thermal energy

Power is energy transferred per unit time. A force's instantaneous mechanical power is F · v.

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v} \tag{52}$$

For steady motion against a drag force of magnitude kv^2, the required mechanical power is kv^3. Doubling speed then requires eight times the power devoted to that drag. Actual vehicles also face rolling resistance, drivetrain losses, and speed-dependent changes in drag behaviour; the cube rule is conditional.

5.9. Useful power and input energy

Lift a 70 kg load through 500 m in 20 minutes, using the constant-g approximation. The useful increase in gravitational energy is 343 kJ, and average useful mechanical power is about 286 W.

If the energy converter in this hypothetical example has efficiency $\eta = 0.25$, input energy is $E_{useful}/\eta = 1.37$ MJ. The efficiency is a stated model parameter, not an estimate of an individual's dietary requirements.

Mechanical work can also be transferred into thermal energy. For a material whose specific heat c_p is approximately constant over the stated temperature interval, heating a mass m through $\Delta\Theta$ requires $Q = mc_p\Delta\Theta$. Here c_p denotes heat capacity per unit mass and is unrelated to the speed of light c.

One small calorie is 4.184 J; one food Calorie is one kilocalorie, or 4,184 J. Keeping those units distinct prevents a factor-of-one-thousand error.

6. Keplerian orbits and orbital rendezvous

6.1. A meeting set by orbital time

Two astronauts share a circular orbit, with Mary ahead of Peter. Peter sends a sandwich backward relative to his own motion while its Earth-centred velocity remains forward. The reduced initial speed places it on an inward-reaching ellipse with a shorter period. Whether it meets Mary depends on the time both take to return to the release point.

The local impulse and the eventual meeting belong to different parts of the trajectory. To relate them, we need the geometry, energy and angular momentum of the complete orbit.

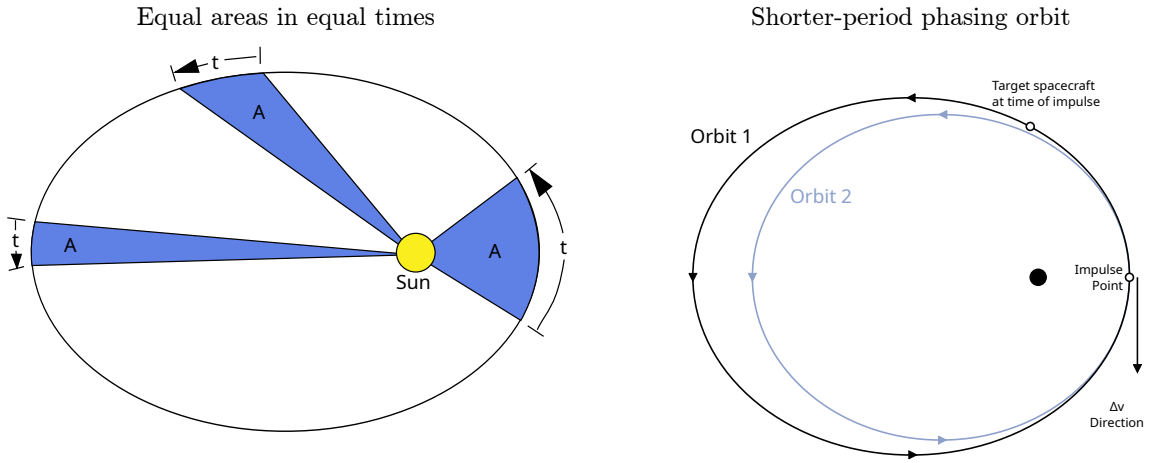


Figure 9. Orbital diagrams. Left: Kepler's second law, with three equal areas swept in equal time intervals. Vector illustration by RJHall, adapted by Talifero, CC BY-SA 2.0 Austria [21]. Right: a smaller-semimajor-axis phasing orbit, reproduced from Bryan Weber, figure 68, CC BY-SA 4.0 [18]. Orbit 2 has the shorter period, with a retrograde impulse at the shared apoapsis. The source illustrates two ellipses sharing the occupied focus schematically; it is not a scale drawing of the circular-orbit numerical example in the text.

6.2. Geometry of the ellipse

For closest radius r_p and farthest radius r_a, the semimajor axis is their average. Eccentricity e describes departure from a circle.

$$a = \frac{r_p + r_a}{2}, \qquad e = \frac{r_a - r_p}{r_a + r_p} \tag{53}$$

Equivalently, r_p = a(1−e) and r_a = a(1+e). A circle has e = 0; a nondegenerate bound ellipse has 0 < e < 1. Both distances are measured from the occupied focus, not from the ellipse's centre.

The conic form of an inverse-square orbit is supplied here. The calculation that follows derives its energy, angular momentum and period relations from that shape and the conservation laws.

6.3. Areal velocity and angular momentum

Over a short time dt, a radius vector and the small displacement form a thin triangle. Its area is one-half the base times perpendicular height. If θ is the angle between r and v:

$$\frac{dA}{dt} = \frac{1}{2} r v \sin\theta = \frac{h}{2}, \qquad h = \frac{L}{m} \tag{54}$$

Central gravity has zero torque about the occupied focus, so h is constant and equal times sweep equal areas. At the apses, where radial speed vanishes, this gives $r_p v_p = r_a v_a$. The smaller apsidal radius is accompanied by the larger speed.

Angular momentum fixes the plane as well: the constant nonzero vector L is perpendicular to both r and v. The ideal orbit stays in the plane normal to L.

6.4. Energy and semimajor axis

For a bound Kepler ellipse, energy depends on a rather than on eccentricity:

$$E = -\frac{\mu m}{2a} \tag{55}$$

Combine this with E = $mv^2/2 - \mu$m/r to get the vis-viva equation.

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \tag{56}$$

At a fixed radius, more total energy means a larger speed. At a fixed a, the total energy and period are the same for ellipses with different eccentricities. Their angular momentum, closest approach, and speed distribution need not be the same.

These formulas use a > 0 for bound ellipses. A signed semimajor-axis convention can extend energy formulas to hyperbolas, but it is not needed in this paper.

6.5. Energy at the orbital turning points

At an apsis, radial speed is zero and transverse speed is h/r. Equate specific energies ε = E/m at the two apses.

$$\frac{h^2}{2r_p^2} - \frac{\mu}{r_p} = \frac{h^2}{2r_a^2} - \frac{\mu}{r_a} \tag{57}$$

For a noncircular ellipse, factoring the differences allows $r_a - r_p$ to cancel. This gives $h^2 = 2\mu r_p r_a/(r_p + r_a)$. Substituting $r_p r_a = a^2(1 - e^2)$ yields

$$h^2 = \mu a(1 - e^2) \tag{58}$$

Substitute that h^2 into the energy at r_p. Using r_p = a(1−e), the factors simplify to $\varepsilon = -\mu/(2a)$. A circle follows as the e → 0 limit. This derivation uses algebra and conservation laws, while taking the elliptical shape as given.

6.6. Derivation of the orbital period

The area of the ellipse is πab, with semiminor axis b = a$\sqrt{1-e^2}$. Divide its full area by the constant area rate h/2.

$$T = \frac{2\pi ab}{h} = 2\pi\sqrt{\frac{a^3}{\mu}} \tag{59}$$

The factors involving eccentricity cancel. For a fixed central mass, all bound ellipses of the same semimajor axis consequently have the same period. The geometrical area and the conserved area rate together establish this result within the supplied Kepler model.

6.7. Recovery of orbital elements from initial data

Suppose r_0, v_0, the angle ϕ_0 between them, and μ are given. First calculate specific energy ε and specific angular momentum h.

$$\varepsilon = \frac{v_0^2}{2} - \frac{\mu}{r_0}, \qquad h = r_0 v_0 \sin\phi_0 \tag{60}$$

For a bound nonradial ellipse, determine a from energy, then e from h. The two apses and the period follow.

$$a = -\frac{\mu}{2\varepsilon}, \qquad e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}} \tag{61}$$

The sign of the initial radial velocity distinguishes inward from outward motion. The scalar values above do not by themselves specify the orbit's full orientation in three-dimensional space.

6.8. Reconstructing an orbit from a tangential velocity

At radius 8.00 × 10^6 m around Earth, give a spacecraft a purely tangential speed of 7,000 m s^{-1}. Compute:

$\varepsilon = -2.5325 \times 10^7$ J kg^{-1}, h = 5.60 × 10^{10} m^2 s^{-1}, a = 7.8697 × 10^6 m, and e = 0.01656.

Thus $r_p = 7.7394 \times 10^6$ m and $r_a = 8.0000 \times 10^6$ m. The initial point is apoapsis because the chosen speed is below the local circular value. The period is about 115.8 minutes.

The tangential initial velocity requires the initial point to be an apsis. Recovering the supplied 8.00 × 10^6 m as the farthest radius checks that identification. The small eccentricity is consistent with the small departure from circular speed.

6.9. Effect of an impulsive velocity change

At a point on an initial circular orbit, a brief prograde tangential impulse increases energy and angular momentum. If the new orbit is still bound, that point becomes periapsis of an ellipse extending outward. Its period is longer.

A brief retrograde impulse that reduces the speed without reversing it makes that point apoapsis of an inward-reaching ellipse. Its semimajor axis and period are smaller.

This is the phasing mechanism: the smaller speed immediately after the impulse produces an orbit with a smaller semimajor axis and a shorter period. "Throw backward" describes the change relative to the astronaut; the sandwich's Earth-centred velocity can remain prograde.

6.10. One-revolution phasing

Let the original circular radius be R and let Mary be ahead by a fraction f of one full circuit. If Peter throws at point X, Mary returns to X after $(1-\mathrm{f})T_c$. Choose a sandwich ellipse that completes one circuit in exactly that time.

$$T_s = (1-f)T_c, \qquad a_s = R(1-f)^{2/3} \tag{62}$$

The period law determines a_s. For the intended nondegenerate inward ellipse, $R/2 < a_s < R$, so X is its apoapsis and its periapsis is positive. The chosen separation must permit that geometry. Vis-viva then determines the speed at X.

$$v_s = \sqrt{\mu\left(\frac{2}{R} - \frac{1}{a_s}\right)}, \qquad \Delta v = v_s - v_c \tag{63}$$

For R = 7,000 km and f = 0.05, the astronauts' period is 97.14 minutes. The sandwich period is 92.28 minutes, its semimajor axis is 6,764.7 km, and its speed at X is 7.4136 km s^{-1}, compared with 7.5461 km s^{-1} for Peter.

Peter's relative throw must therefore be about 132 m s^{-1} backward. This is a mathematical delivery mechanism, not a plausible unassisted hand throw.

The timing condition produces an encounter, not yet a completed rendezvous. On returning to X, the sandwich still has speed v_s, while Mary has circular speed v_c. Their relative speed is therefore the same 132 m s^{-1}. A second, prograde impulse of magnitude $v_c - v_s$ would match their velocities in this ideal model. [18]

The periapsis is $2a_s - R = 6,529.4$ km, about 158 km above the chosen mean Earth radius. Atmospheric drag would matter there for a real object. Together with the required throw speed, this makes the result an illustration of ideal two-body timing rather than a feasible sandwich delivery.

6.11. Phasing over several revolutions

Suppose the sandwich completes n orbits while Mary travels k−f full circuits before reaching X, for integers n and k. Then:

$$nT_s = (k-f)T_c, \qquad \frac{a_s}{R} = \left(\frac{k-f}{n}\right)^{2/3} \tag{64}$$

Taking k = n makes $T_s/T_c = 1-\mathrm{f/n}$, closer to 1 as n grows. The required retrograde impulse can be smaller, at the price of a longer wait. Both the geometry and physical clearance still need checking.

The example illustrates a general phasing calculation: the meeting event fixes a timing condition, which in turn constrains the period and semimajor axis.

7. Gravitational inference from orbital motion

7.1. Inferring the source from the trajectory

The preceding sections prescribe a source and calculate a trajectory. Observed motion allows the direction of the calculation to be reversed. A radius and a period can constrain the source mass, provided the force law and orbital geometry are specified. The assumptions used to predict the orbit therefore also govern the inference.

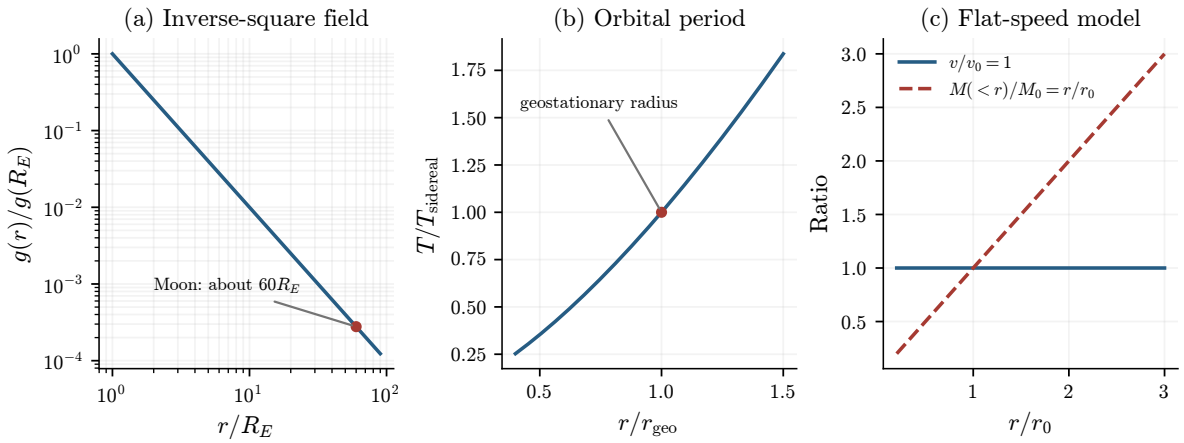


Figure 10. Calculated gravitational relations. (a) Inverse-square acceleration, with the approximate lunar radius marked. (b) Keplerian period normalised to the geostationary radius and sidereal day. (c) A hypothetical flat circular-speed curve and inferred enclosed mass in a spherical model, with $M_0 = v_0^2 r_0/G$. These retained graphs are model calculations, not measured data.

7.2. Comparison of terrestrial and lunar acceleration

Near Earth's surface an apple accelerates at about g. The Moon is roughly 60 Earth radii from Earth's centre. If gravity follows an inverse-square law, its acceleration there is about $\mathrm{g}/60^2 = \mathrm{g}/3{,}600 \simeq 0.00273$ m s^{-2}.

Independently estimate the Moon's circular-orbit acceleration from its measured period T_M and radius r_M.

$$a_M = \frac{4\pi^2 r_M}{T_M^2} \tag{65}$$

With $r_M \simeq 3.84 \times 10^8$ m and $T_M \simeq 27.3$ days, the result is about 0.00272 m s^{-2}. The agreement connects a falling object near Earth with a continually falling object far away. The Moon's real orbit is not exactly circular, so this is deliberately a first approximation.

7.3. Inference of the force exponent

For circular motion, acceleration is proportional to r/T^2. Suppose the acceleration supplied by a central force varies as $1/r^n$. Equating the scalings gives T^2 proportional to r^{n+1}.

$$a_c = \frac{4\pi^2 r}{T^2}, \qquad a_c \propto r^{-n} \implies T^2 \propto r^{n+1} \tag{66}$$

Kepler's T^2 proportional to r^3 therefore points to n = 2. The proportionality constant still needs physical interpretation. Newton's full law includes both interacting masses and G; orbital data for a small test body determine GM directly.

7.4. Determination of a central mass

Rearrange the circular period relation.

$$M = \frac{4\pi^2 r^3}{GT^2} \tag{67}$$

The distance is from centre to centre in the spherical-body approximation. If the orbiting body's mass is not negligible, the relative two-body version contains M + m rather than M alone. This paper normally assumes m $\ll$ M.

The shell theorem allows a spherically symmetric body's external gravitational field to be treated as though all its mass were at its centre. That is a supplied result here, not a claim that all extended mass distributions have point-mass fields. Inside such a body, use an appropriate interior model.

7.5. The radius of a geostationary orbit

One sidereal day is approximately 86,164 s. Solve the circular period formula for r using that period and Earth's μ.

$$r = \left[\mu \left(\frac{T}{2\pi}\right)^2\right]^{1/3} \simeq 42,164 \text{ km} \tag{68}$$

Subtract the chosen Earth radius to get an altitude of about 35,793 km. The often quoted 35,786 km uses a more appropriate equatorial radius; the difference is a radius convention, not a new orbital law.

Matching the rotation period alone makes an orbit geosynchronous. To remain over one fixed ground point in the ideal model, it must also be circular, equatorial, and prograde. An inclined or eccentric geosynchronous orbit generally wanders in the ground observer's sky.

7.6. Potential references in orbital mechanics

The reference choices can be reconciled by comparing mgz near the surface with −GMm/r for large distances. They need not have the same numerical zero. Their changes agree in the small-height limit once the same reference is used.

For escape classification, the convention $U(\infty) = 0$ gives the sign of E a particularly useful meaning. If U is shifted by a constant, the physical escape condition is unchanged, provided the threshold energy is shifted by that same constant. "Negative means bound" is tied to the stated reference.

7.7. A hypothetical flat rotation curve

Under an approximately spherical mass model, a circular tracer at radius r and speed v implies:

$$M(< r) = \frac{v^2 r}{G} \tag{69}$$

If v stays roughly constant as r increases, inferred enclosed mass grows roughly in proportion to r. If nearly all gravitating mass had already been enclosed, one would instead expect v proportional to $1/\sqrt{r}$.

For an illustration, a constant 200 km s^{-1} at 10 kpc gives about 9.3×10^{10} solar masses enclosed in this spherical approximation. At 20 kpc with the same speed, it gives twice that amount. The numbers are a hypothetical example, not measurements of a named galaxy.

A disk or a noncircular tracer requires a different calculation. Rotation curves contribute evidence for unseen gravitating matter only through such modelling. The hypothetical spherical estimate above neither identifies a particle species nor supplies a summary of cosmological evidence.

8. Proper time and Schwarzschild geometry

8.1. From orbital timing to clock comparison

In the Newtonian calculation, a period is measured against a common time coordinate. Inferring a source from that period does not require a law for how the observer's clock responds to gravity. A comparison between clocks at different heights, or between a supported observer and a falling observer, asks for information that this framework does not contain.

General relativity changes the description of free motion. An unforced test body follows a geodesic of spacetime, while a supported body is kept from that motion by a nongravitational force. A sufficiently small freely falling laboratory admits a local inertial description. Over an extended region, relative free-fall accelerations can reveal tidal curvature. Neither a change of coordinates nor the removal of a local support force eliminates those tidal effects. [8, 9]

The interval relates coordinate changes to distances and clock readings. A trajectory must now be accompanied by a specification of whose time is being calculated. This becomes decisive near a black-hole horizon, where a finite falling-clock interval and an unbounded Schwarzschild time can describe the approach to the same boundary. We first establish the clock distinction in flat spacetime, then use the supplied Schwarzschild exterior geometry.

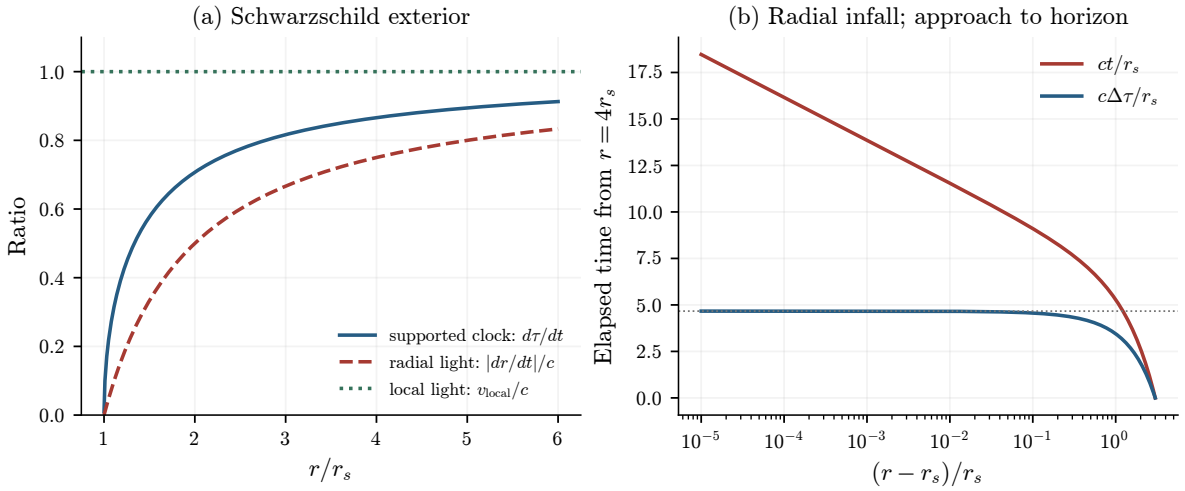


Figure 11. Calculated Schwarzschild relations. (a) Supported-clock rate, radial coordinate light speed and locally measured light speed outside the horizon. (b) Coordinate and proper time during radial infall from rest at infinity, with time origins at $r = 4r_s$. The proper-time limit is $14r_s/(3c)$, whereas the coordinate time diverges.

8.2. Proper time and the flat spacetime interval

An event is a location at a time. In flat spacetime, two nearby events along a traveller's path satisfy the interval relation below, using a plus-time sign convention.

$$c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \tag{70}$$

The coordinate time t belongs to the chosen inertial description. Proper time τ belongs to a clock travelling along the worldline. If the traveller's speed in that frame is v, divide by dt^2.

$$d\tau = dt\sqrt{1 - \frac{v^2}{c^2}} \tag{71}$$

For variable speed, integrate that expression over t. All changing quantities are functions of the single variable t. This is ordinary one-variable calculus.

For timelike separation, c^2dt^2 is greater than the squared spatial displacement; a massive traveller can connect the nearby events. Lightlike separation has equality, and light follows the boundary. Spacelike-separated events are too far apart in space for a light signal to connect them in that time interval.

A null path has zero spacetime interval. It has no associated photon rest frame or carried clock, so it does not define a photon's experienced time.

8.3. Elapsed time on a moving clock

A clock moves at a constant 0.6c for a coordinate interval of 10 s in an inertial frame. Its elapsed time is $10\sqrt{1-0.36} = 8$ s.

The 8 s belongs to the specified segment of the moving clock's worldline. A complete reunion comparison requires both complete paths. Different elapsed times can result from different worldlines while both clocks function normally.

8.4. Coordinates and metric coefficients

Even ordinary flat geometry can look different in different coordinates. In a plane, a short radial step dr and a short angular step dϕ have squared distance $dr^2 + r^2d\phi^2$. The factor r^2 accounts for the increase in arc length at larger radius for the same angular change.

Metric coefficients specify how coordinate changes enter the interval. Their coordinate dependence alone does not establish curvature, as the polar-coordinate example already shows. Curvature concerns the geometry itself, including the relative behaviour of neighbouring freely falling paths.

The geodesic description concerns bodies with no nongravitational force acting on them. A floor holding a body at rest is an additional physical constraint. Special relativity applies locally in a sufficiently small freely falling laboratory, with tidal effects limiting how far that approximation can be extended. [8]

The corresponding variational statement for a massive particle is stationary proper time. On sufficiently short timelike segments, the free path locally maximises elapsed proper time relative to nearby paths joining the same events. This result is supplied without a variational derivation. It is not a global rule that every geodesic always maximises or minimises time. [9]

8.5. The Schwarzschild exterior interval

Take the exterior of a spherical, nonrotating, uncharged source in the ideal static setting. Choose a plane through the centre for the motion; spherical symmetry lets us study that plane. Define the Schwarzschild radius r_s and a convenient factor A(r).

$$r_s = \frac{2GM}{c^2}, \qquad A(r) = 1 - \frac{r_s}{r} \tag{72}$$

The interval for motion in that plane is:

$$d\tau^2 = A\,dt^2 - \frac{dr^2}{c^2 A} - \frac{r^2 d\phi^2}{c^2} \tag{73}$$

Here r is the circumferential radius: a symmetry circle at fixed r has circumference 2πr. It is not automatically the proper radial distance measured by laying down local rulers. Coordinate t is normalised so stationary clocks very far away approach its rate.

The interval is supplied as the Schwarzschild vacuum exterior solution for the stated ideal source. Its exactness belongs to that model. The field equations needed to derive it are outside this manuscript. Agreement of one coefficient with a Newtonian limit does not by itself determine the remaining coefficients.

The vacuum formula does not describe the material interior of Earth. Its formal r_s lies outside its domain of application there, so it cannot be used to infer a horizon inside the planet. Matter requires an appropriate interior solution.

8.6. The Newtonian weak-field limit

For the exterior Newtonian potential $\Phi = -GM/r$, the Schwarzschild factor is $A = 1 + 2\Phi/c^2$. When $|\Phi|/c^2$ is small, the time coefficient is close to one.

A stationary clock has $dr = d\phi = 0$, giving $d\tau/dt = \sqrt{A}$. Applying $\sqrt{1+x} \simeq 1 + x/2$ for $|x| \ll 1$ gives

$$\frac{d\tau}{dt} \simeq 1 + \frac{\Phi}{c^2} \tag{74}$$

A more negative potential gives a smaller supported-clock rate relative to the distant time coordinate. The weak-field relation connects an earlier calculation of potential to a new observable, a clock-rate comparison. It does not identify the Newtonian potential with spacetime curvature.

If motion is also slow, the leading correction includes the familiar special-relativistic term $-v^2/(2c^2)$. Keeping those leading terms is an approximation, not an exact formula for arbitrary strong gravity and motion.

8.7. Two clocks held at fixed radii

Hold one clock stationary at r $= 2r_s$ and another at r $= 4r_s$. They are outside the horizon and have no angular motion. Their rates relative to t are $\sqrt{1/2} \simeq 0.707$ and $\sqrt{3/4} \simeq 0.866$.

During the same coordinate interval, the inner clock accumulates $0.707/0.866 \simeq 0.816$ times the outer clock's elapsed time. Both clocks are supported. Their fixed positions, rather than a free-fall condition, justify the rate formula used here.

At the horizon, an observer cannot remain stationary in this way. Taking the stationary expression toward zero does not describe a physical clock hovering exactly on the horizon.

8.8. Coordinate and local light speeds

For radial light, $\mathrm{d}\phi = 0$ and $d\tau^2 = 0$. Substitute into the metric, rearrange, and take the two square-root signs for outgoing and ingoing rays.

$$\frac{dr}{dt} = \pm cA = \pm c\left(1 - \frac{r_s}{r}\right) \tag{75}$$

This coordinate speed approaches zero as r approaches r_s from outside. The numerator is a change in circumferential coordinate r; the denominator is the distant time coordinate t. Neither is automatically a local ruler-and-clock reading.

For a stationary observer at $r > r_s$, a small proper radial length is $d\ell = |dr|/\sqrt{A}$ and the local clock interval is $dt_{\mathrm{local}} = \sqrt{A}\,dt$. Both conversions are needed to obtain a locally measured speed.

$$\frac{d\ell}{dt_{\mathrm{local}}} = \frac{|dr/dt|}{A} = c \tag{76}$$

The local measurement gives c despite the radius dependence of dr/dt. Freely falling local inertial observers also measure c. The supported observer used for this conversion exists only outside the horizon. [8, 10]

8.9. Coordinate time for radial light

For an ingoing radial light ray, $\mathrm{dt/dr} = -1/(\mathrm{cA})$. Starting from $r_0 > r_s$ and reaching r with $r_s < \mathrm{r} < r_0$, integrate using $1/\mathrm{A} = \mathrm{r}/(\mathrm{r}-r_s)$.

$$\Delta t = \frac{r_0 - r}{c} + \frac{r_s}{c}\ln\left(\frac{r_0 - r_s}{r - r_s}\right) \tag{77}$$

The logarithm diverges as r approaches r_s from above. Schwarzschild t consequently becomes unbounded along this ingoing ray. The divergence belongs to this coordinate chart at the horizon. The local speed calculation has already given the finite value c.

A divergence of a coordinate quantity need not identify a physical singularity. The behaviour of locally measurable quantities must be examined independently.

8.10. Proper time during radial infall

To make the comparison explicit, use a freely falling massive test particle on a radial trajectory that started at rest at infinity. It is already moving when it passes any finite r_0. Its conserved relativistic energy gives $\mathrm{A\ dt/d}\tau = 1$. This conservation relation is supplied at the present level.

Divide the radial interval by $d\tau^2$, use $\mathrm{dt/d}\tau = 1/\mathrm{A}$, and solve for the inward branch. This gives:

$$\frac{dr}{d\tau} = -c\sqrt{\frac{r_s}{r}}, \qquad \frac{dr}{dt} = -cA\sqrt{\frac{r_s}{r}} \tag{78}$$

The proper-time radial derivative stays finite at the horizon while the coordinate derivative tends to zero. The value $\mathrm{dr}/\mathrm{d}\tau$ is not itself a local speed measured relative to a nearby observer; r is still the chosen radial coordinate.

Integrating the first relation from r_0 down to r_s requires only a one-variable integral of $\sqrt{r}$.

$$\Delta\tau = \frac{2r_s}{3c}\left[\left(\frac{r_0}{r_s}\right)^{3/2} - 1\right] \tag{79}$$

From $r_0 = 4r_s$, the falling clock accumulates $14r_s/(3c)$ before reaching the horizon. The initial condition remains rest at infinity. The object is already moving at $4r_s$.

For this same trajectory, the second relation gives $dt/dr = -\sqrt{r/r_s}/(cA)$. Near the horizon, $A = (r - r_s)/r$ and $\sqrt{r/r_s}$ tends to one. The leading integral is therefore logarithmic, so Schwarzschild time diverges while the proper-time integral stays finite. This conclusion uses the supplied geodesic energy constant. Regular continuation through the horizon is a property of the black-hole geometry, not something established by extending the static coordinate chart.

8.11. Signals received by a distant observer

A coordinate description is not the image received at a telescope. Signals emitted progressively nearer the horizon can arrive with increasing delay and redshift, causing the source to fade. The statement that an observer sees a bright object frozen forever omits this signal propagation.

The horizon marks a causal boundary in the ideal black-hole spacetime. It is not a solid material wall. Inside the black-hole region, future-directed causal motion cannot escape to the exterior. A full treatment of the interior needs coordinates that remain regular at the horizon; replacing them is beyond the mathematical scope chosen for this paper.

8.12. Horizon radius and the central singularity

The expression $r_s = 2\mathrm{GM}/c^2$ gives about 2.95 km per solar mass. A nonrotating 10-solar-mass black hole therefore has a horizon radius of about 29.5 km. The horizon is distinct from the classical singularity at r = 0.

Setting Newtonian escape speed equal to c produces the same algebraic radius. That coincidence is a useful memory aid, not a derivation of black holes: Newtonian mechanics does not provide the causal structure, clock behaviour, or complete Schwarzschild geometry.

9. Selection and application of physical methods

The unknown determines which relation is useful. A local support force, an orbital period and a clock reading are different observables even when they refer to one physical

system. Table 2 records the assumptions that must be checked before applying the corresponding calculation.

Table 2: Selection of the relevant physical method.

Required quantity	**First tool to try**	**What to check**
Force or support at one point on a curve	Net inward force = mv^2/r	Direction toward centre; other real forces
Speed after moving between positions	Work and energy	Which forces do work; potential reference
Spin change when mass moves	Angular momentum	External torque and chosen axis
Shape and timing of an orbit	Specific energy plus specific angular momentum	Two-body approximation and collision clearance
Mass inferred from a tracer	Circular dynamics or Kepler's period law	Geometry and whether the tracer is circular
A clock comparison	Interval or proper-time relation	Which worldline and which time coordinate

9.1. Dependence on the assumptions

Changing an assumption tests the reasoning before any new calculation is made. The following changes affect distinct parts of the model.

- Remove string tension at one point on a circular path. The initial velocity survives; the circular trajectory generally does not.
- A radial impulse has zero angular impulse about the centre. It can change the energy, although a specially chosen impulse may leave the speed magnitude unchanged.
- A nonzero tangential impulse changes angular momentum about the centre and generally changes energy. A reversal with the same speed magnitude is an exception for the energy.
- Shift the zero of U. Forces and energy differences survive; the numerical escape threshold shifts with the convention.
- Change a distant coordinate description to a local clock measurement. A coordinate speed can change; the locally measured light speed remains c.

10. Principal equations and terminology

Table 3: Principal relations and their domains of validity.

Idea	**Relation**	**Conditions**
Circular speed and period	$\mathrm{v} = \omega\mathrm{r} = 2\pi\mathrm{r}/\mathrm{T}$	Circular path
Inward acceleration	$a_c = v^2/\mathrm{r} = \omega^2\mathrm{r}$	Radius of curvature is r
Particle angular momentum	$\mathrm{L} = \mathrm{r} \times \mathrm{mv}$	Specified origin and inertial frame
Torque law	$\mathrm{dL}/\mathrm{dt} = \tau_{external}$	Fixed inertial origin; internal torque cancellation
Axial rigid rotation	$\mathrm{L} = \mathrm{I}\omega$, $\mathrm{K} = I\omega^2/2$	Appropriate symmetry or principal axis
Work energy	$W_{net} = \Delta\mathrm{K}$	Classical particle mechanics
Gravitational potential energy	$\mathrm{U} = -\mu\mathrm{m}/\mathrm{r}$	Spherical exterior; zero at infinity
Gravitational potential	$\Phi = U/m = -GM/r$	Spherical exterior; zero at infinity
Field from potential	$g_r = -d\Phi/dr$	Signed outward radial component
Field at altitude	$g(h) = g_0/(1 + h/R)^2$	Exterior spherical source; $g_0 = GM/R^2$
Field at depth	$g(d) = g_0(1 - d/R)$	Uniform density only; $0 \le d \le R$
Interior potential	$\Phi = -GM(3R^2 - r^2)/(2R^3)$	Uniform sphere; $r \le R$; infinity reference
Circular orbit	$v_c = \sqrt{\mu/r}$, $\mathrm{T} = 2\pi\sqrt{r^3/\mu}$	Dominant central mass
Escape threshold	$v_{esc} = \sqrt{2\mu/r}$	Isolated coast model; unobstructed outward escape
Kepler ellipse	$\mathrm{E} = -\mu\mathrm{m}/(2\mathrm{a})$, $h^2 = \mu\mathrm{a}(1-e^2)$	Bound nonradial inverse-square orbit
Vis viva	$v^2 = \mu(2/\mathrm{r}-1/\mathrm{a})$	Same two-body ellipse
Area rate	$\mathrm{dA}/\mathrm{dt} = \mathrm{h}/2$	Central force about the chosen centre
Power	$\mathrm{P} = \mathrm{F} \cdot \mathrm{v}$	Instantaneous mechanical transfer
Flat proper time	$\mathrm{d}\tau = \mathrm{dt}\sqrt{1 - v^2/c^2}$	Inertial coordinate frame
Schwarzschild factor	$\mathrm{A} = 1-r_s/\mathrm{r}$, $r_s = 2\mathrm{GM}/c^2$	Ideal spherical exterior model
Supported clock	$\mathrm{d}\tau/\mathrm{dt} = \sqrt{A}$	Stationary outside horizon
Radial coordinate light speed	$\mathrm{dr}/\mathrm{dt} = \pm\mathrm{cA}$	Schwarzschild t and r; radial light

10.1. Terminology

Centripetal means directed toward the centre of curvature. **Normal force** is the contact force perpendicular to a surface. **Torque** is the rate of change of angular momentum about the same origin. **Moment of inertia** measures mass distribution

relative to a chosen rotation axis.

Conservative means the force's work between two points is independent of the path in the domain considered. **Gravitational potential** is gravitational potential energy per unit test mass, expressed in joules per kilogram. **Specific energy** is energy per unit test mass. **Specific angular momentum** is angular momentum per unit test mass. **Periapsis** is closest approach; **apoapsis** is farthest approach.

Metric is the rule for calculating intervals from coordinate changes. **Proper time** is elapsed time along a clock's worldline. **Geodesic** is the free-motion path in the geometry. **Horizon** is a causal boundary, rather than a material surface or the central singularity.

11. Numerical consistency of a Kepler orbit

The analytic orbit relations can check a numerical implementation of the same force law. Choose dimensionless units with $GM = a = 1$, initial position $(0.4, 0)$ and initial velocity $(0, 2)$. These give eccentricity 0.6, specific energy -0.5, specific angular momentum 0.8 and analytic period 2π. The force law is prescribed in both calculations.

The inverse-square acceleration is integrated using the velocity-Verlet method. With position vector $\mathbf{r}_n$, velocity $\mathbf{v}_n$, acceleration $\mathbf{a}_n = -\mathbf{r}_n/|\mathbf{r}_n|^3$ and fixed time step Δt, the update is

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_n \Delta t + \frac{1}{2}\mathbf{a}_n (\Delta t)^2 \tag{80}$$

$$\mathbf{a}_{n+1} = -\frac{\mathbf{r}_{n+1}}{|\mathbf{r}_{n+1}|^3}, \qquad \mathbf{v}_{n+1} = \mathbf{v}_n + \frac{1}{2}(\mathbf{a}_n + \mathbf{a}_{n+1})\Delta t \tag{81}$$

The principal run uses 20,000 steps per analytic period. Specific energy and specific angular momentum are recalculated at every step. Their conservation is not imposed by a separate correction. The maximum relative energy error is 7.31×10^{-7} and the maximum relative angular-momentum error is 1.94×10^{-14}. After one analytic period the position error is 1.35×10^{-5} in semimajor-axis units.

Runs with 10,000 and 40,000 steps per period bracket the principal calculation. Over this fixed one-period interval, each halving of the time step reduces the maximum relative energy error by approximately a factor of four. This is consistent with second-order convergence for this case. The comparison does not establish the same error behaviour for every orbit or for arbitrarily long integration times.

The graphs are calculations in Python. The conceptual diagrams retain the attribution of their sources. Machine-readable trajectories, parameter records and integration code accompany the manuscript. Agreement with the analytic ellipse and the step-size comparison check this implementation of the chosen model. They supply no new astronomical measurements and do not independently validate the inverse-square law.

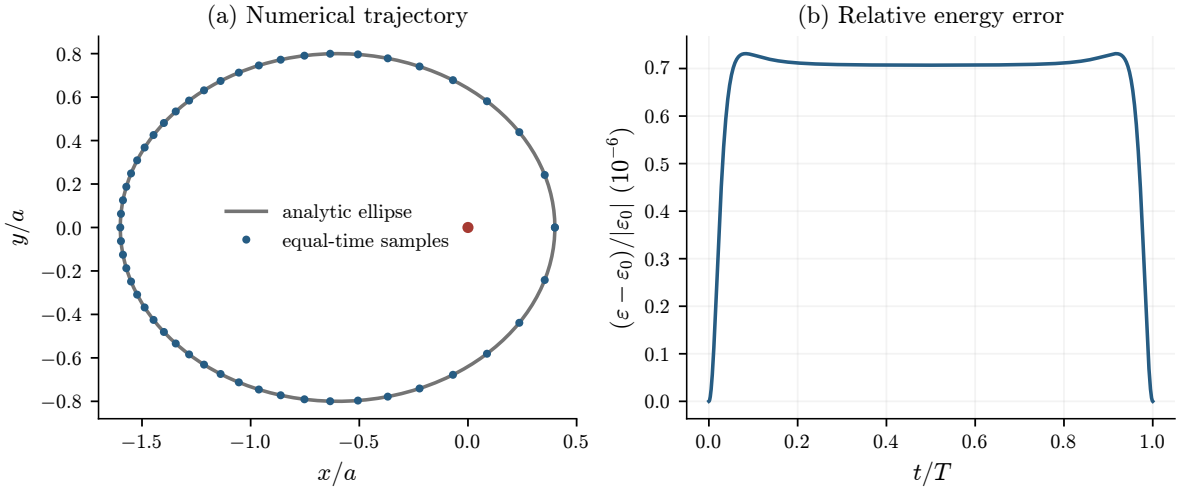


Figure 12. Velocity-Verlet integration of an eccentricity-0.6 Kepler orbit. (a) Equal-time numerical samples overlaid on the analytic ellipse. Their greater density near apoapsis reflects the slower orbital motion there. (b) The relative specific-energy error over one analytic period. All points are synthetic calculations, rather than measured astronomical data.

12. Conclusions and scope

The calculation of gravitational motion begins with a vector state and a model of the source. Circular acceleration specifies the required inward force. Under a central force, angular momentum is conserved about the centre. Work and energy then constrain the speed and the accessible radii. Taking the inverse-square conic shape as given, the turning-point calculation yields the relation between energy and semimajor axis. The area rate supplies the period. Phasing satisfies a meeting-time condition; completing a rendezvous also requires matching velocities.

For spherical sources, the altitude law follows without a uniform-density assumption. The interior field instead depends on enclosed mass. Uniform density gives a linear depth law and a quadratic interior potential, matched to the exterior at the surface. The nonuniform formula shows why an actual density profile can reverse the near-surface trend. Zero central field expresses zero potential slope and leaves the infinity-referenced potential finite and negative.

The numerical orbit reproduces the prescribed Kepler motion within the reported errors. Over one analytic period, the step-size comparisons are consistent with second-order energy-error convergence. This checks the numerical calculation against the analytic model. Observational confirmation would require measured quantities and an assessment of their uncertainties, neither of which is provided by these synthetic data.

The relativistic calculations use the Schwarzschild exterior interval and a supplied radial geodesic energy constant. They distinguish supported-clock readings from falling-clock readings and both from coordinate time. The local light speed remains c, while a radial Schwarzschild coordinate speed tends to zero at the horizon. For the specified

infall, the falling clock reaches that boundary in finite proper time and the Schwarzschild time diverges. The different time limits follow from the different quantities being integrated.

The analysis is restricted to test-particle motion, specified spherical density models and the ideal Schwarzschild vacuum exterior. It does not derive the field equations or an interior relativistic solution. Drag, source rotation, nonspherical mass distributions, stellar mass loss and significant interaction between orbiting bodies require further modelling. These limits identify what would have to be supplied before applying the calculations to a different physical system.

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